

ContractIQ: Contract Obligation Tracking & Compliance Management Platform

Database Tables Required

1. Users
2. Contracts
3. Contract Versions
4. Obligations
5. Renewals
6. Notifications
7. Reports
8. Audit Logs
9. Activities

Database Schema

Users

: Stores registered users

Column	Data Type	Constraints
id	INTEGER	PK, Auto Increment
full_name	VARCHAR(100)	NOT NULL
email	VARCHAR(255)	UNIQUE, NOT NULL
password	VARCHAR(255)	NOT NULL
role	VARCHAR(50)	NOT NULL
is_active	BOOLEAN	DEFAULT TRUE
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

Primary Key: *id*

Foreign Keys: *None*

Contracts

Stores contract details

Column	Data Type	Constraints
id	INTEGER	PK
title	VARCHAR(255)	NOT NULL
contract_number	VARCHAR(100)	UNIQUE
vendor_name	VARCHAR(255)	NOT NULL
start_date	DATE	NOT NULL
end_date	DATE	NOT NULL
contract_value	DECIMAL(12,2)	
status	VARCHAR(50)	NOT NULL
created_by	INTEGER	FK -> Users.id
created_at	TIMESTAMP	NOT NULL

Primary Key: *id*

Foreign Keys: *created_by -> users.id*

Contract Versions

: Stores versions

Column	Data Type	Constraints
id	INTEGER	PK
contract_id	INTEGER	FK -> Contracts.id
version_number	INTEGER	NOT NULL
file_path	TEXT	NOT NULL
uploaded_by	INTEGER	FK -> Users.id
uploaded_at	TIMESTAMP	NOT NULL

Primary Key: *id*

Foreign Keys: *contract_id -> contracts.id; uploaded_by -> users.id*

Obligations

: Stores obligations

Column	Data Type	Constraints
id	INTEGER	PK
contract_id	INTEGER	FK
title	VARCHAR(255)	NOT NULL
description	TEXT	
due_date	DATE	NOT NULL
priority	VARCHAR(20)	
status	VARCHAR(50)	
assigned_to	INTEGER	FK -> Users.id

Primary Key: *id*

Foreign Keys: *contract_id -> contracts.id; assigned_to -> users.id*

Renewals

: Stores renewals

Column	Data Type	Constraints
id	INTEGER	PK
contract_id	INTEGER	FK
renewal_date	DATE	NOT NULL
reminder_days	INTEGER	
status	VARCHAR(50)	

Primary Key: *id*

Foreign Keys: *contract_id -> contracts.id*

Notifications

: Stores reminders

Column	Data Type	Constraints
id	INTEGER	PK
user_id	INTEGER	FK
contract_id	INTEGER	FK
message	TEXT	NOT NULL
notification_type	VARCHAR(50)	
is_read	BOOLEAN	DEFAULT FALSE
sent_at	TIMESTAMP	

Primary Key: *id*

Foreign Keys: *user_id -> users.id; contract_id -> contracts.id*

Reports

: Stores reports

Column	Data Type	Constraints
id	INTEGER	PK
generated_by	INTEGER	FK
report_name	VARCHAR(255)	
report_type	VARCHAR(100)	
file_path	TEXT	
generated_at	TIMESTAMP	

Primary Key: *id*

Foreign Keys: *generated_by -> users.id*

Audit Logs

: Stores audit history

Column	Data Type	Constraints
id	INTEGER	PK
user_id	INTEGER	FK
action	VARCHAR(255)	
table_name	VARCHAR(100)	
record_id	INTEGER	
action_time	TIMESTAMP	

Primary Key: *id*

Foreign Keys: *user_id -> users.id*

Activities

: Stores user activities

Column	Data Type	Constraints
id	INTEGER	PK
user_id	INTEGER	FK
contract_id	INTEGER	FK
activity	TEXT	
activity_time	TIMESTAMP	

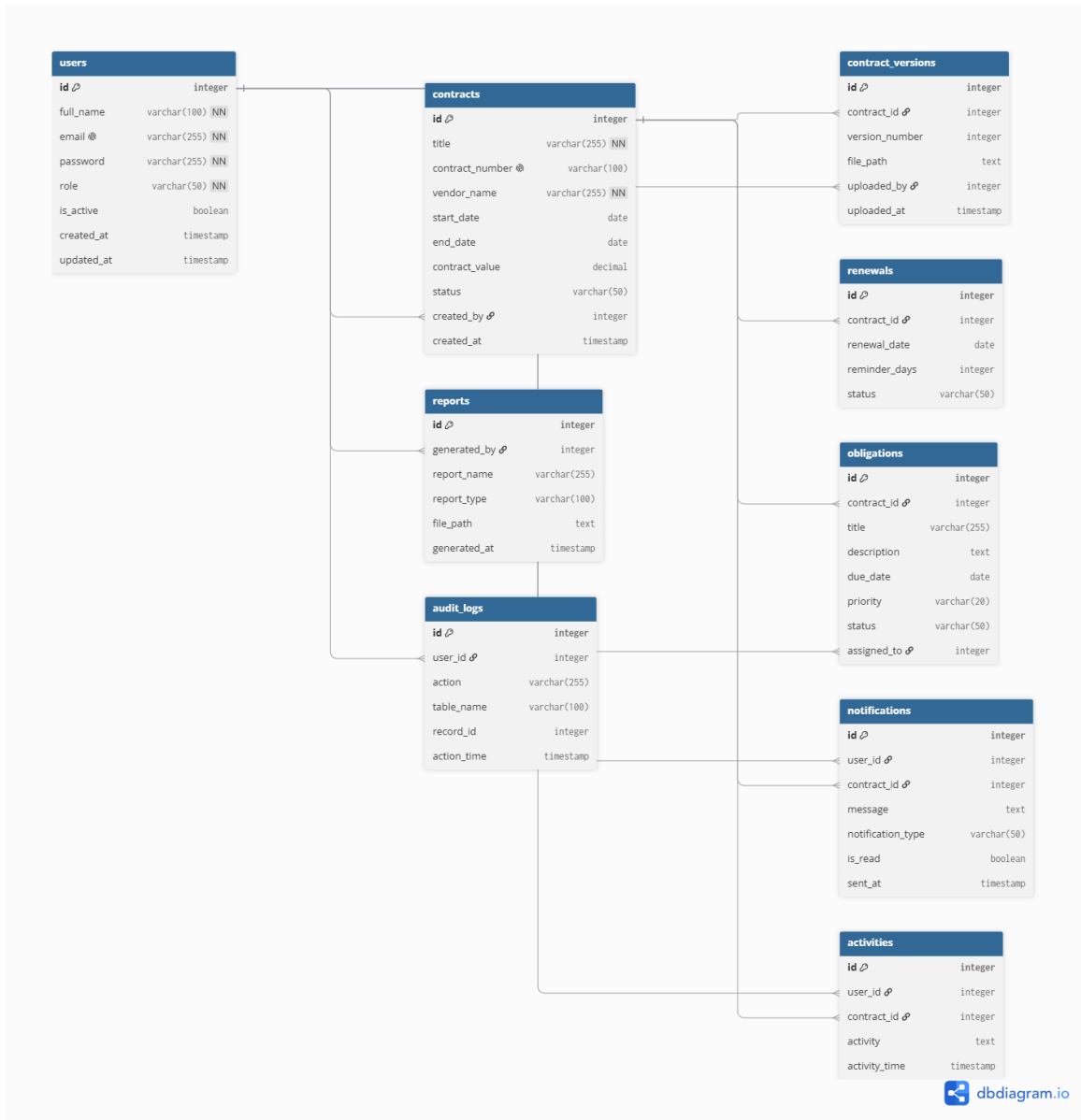
Primary Key: *id*

Foreign Keys: *user_id -> users.id; contract_id -> contracts.id*

Table Relationships

Parent	Child	Relationship
Users	Contracts	One-to-Many
Users	Notifications	One-to-Many
Users	Reports	One-to-Many
Users	Audit Logs	One-to-Many
Users	Activities	One-to-Many
Contracts	Contract Versions	One-to-Many
Contracts	Obligations	One-to-Many
Contracts	Renewals	One-to-One
Contracts	Notifications	One-to-Many
Contracts	Activities	One-to-Many

ER Diagram



How the Tables Work Together

The Users table stores information about all the users who use the ContractIQ system. Users can create and manage contracts. The Contracts table is the main table because it stores all contract details. Each contract can have different versions, obligations (tasks to complete), notifications, activities, and renewal information.

The Reports table stores generated reports, the Audit Logs table keeps a record of important actions performed in the system, and the Activities table records the daily actions of users. All these tables work together to help manage contracts, track important deadlines, maintain records, and ensure compliance.